A First Course in Abstract Algebra

Section 43: More Homology Computations and Applications

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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43 More Homology Computations and Applications

Exercise 43.1: Two triangulations of a square

In the first triangulation there are four corner vertices, four boundary edges and one diagonal, and two triangles. Thus

$$\chi(X) = V - E + F = 4 - 5 + 2 = 1.$$

In the second, the crossing of the diagonals is a vertex, and the horizontal line also introduces a vertex at each midpoint of the left and right sides. Thus there are $4 + 1 + 2 = 7$ vertices. There are six boundary edges and six interior edges (four half-diagonals and two horizontal half-segments), making twelve edges, and six triangles. Hence

$$\chi(X) = 7 - 12 + 6 = 1.$$

Counting intersections and subdivided sides correctly is essential; the two answers agree despite different simplex counts.

Exercise 43.2: Euler characteristic and Betti numbers

The theorem says $\chi = b_0 - b_1 + b_2$ for these two-dimensional spaces, where b_i is the free rank of H_i; torsion contributes no rank. **(a)** The annulus triangulation has five inner and five outer vertices, ten boundary edges and ten other edges, and ten triangles. Thus $\chi = 10 - 20 + 10 = 0$. Its homology has ranks $(1, 1, 0)$, giving $1 - 1 + 0 = 0$. **(b)** In the torus's 3×3 square grid with each small square cut diagonally, opposite sides are identified. This leaves $V = 9$, $E = 27$ (nine horizontal, nine vertical, nine diagonal), and $F = 18$. Thus $\chi = 9 - 27 + 18 = 0$. Its ranks $(1, 2, 1)$ give $1 - 2 + 1 = 0$. **(c)** The same grid with the Klein-bottle edge identifications again gives $V = 9, E = 27, F = 18$, hence $\chi = 0$. Its groups are $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z} \oplus \mathbb{Z}_2$, $H_2 = 0$, so its ranks $(1, 1, 0)$ give $1 - 1 + 0 = 0$. The $\mathbb{Z}_2$ factor does not change the Euler characteristic.

Exercise 43.3: Fixed points on a square and on two disks

For a square the answer is yes: it is homeomorphic to the closed disk E^2, and conjugating a continuous self-map by such a homeomorphism reduces the question to Brouwer's fixed-point theorem. A fixed point of the conjugate map pulls back to one of the original map. For two disjoint disks the answer is no. Choose matching coordinates on the disks and define a map that sends every point of the first to the corresponding point of the second, and conversely. It is continuous on each component, hence on their disjoint union, but has no fixed point because every point changes components. The connected disk hypothesis cannot simply be dropped.

Exercise 43.4: A sphere touching a Klein bottle

The union is connected and is the wedge of a sphere and a Klein bottle at one point. Its positive-dimensional homology is the direct sum of that of the two pieces: cycles on either piece have no new identification except at a vertex. The Klein bottle contributes $\mathbb{Z} \oplus \mathbb{Z}_2$ in dimension one and nothing in dimension two; the sphere contributes $\mathbb{Z}$ in dimension two. Thus

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z} \oplus \mathbb{Z}_2, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

Exercise 43.5: Two disjoint Klein bottles

A disjoint union splits its chain complex into the two component chain complexes. Taking direct sums of $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z} \oplus \mathbb{Z}_2$, $H_2 = 0$ for a Klein bottle gives

$$H_0 \cong \mathbb{Z}^2, \qquad H_1 \cong \mathbb{Z}^2 \oplus \mathbb{Z}_2^2, \qquad H_n = 0 \ (n \geq 2).$$

There are two torsion generators, one from each component, and two H_0 generators because the components are disjoint.

Exercise 43.6: True or false: general homology properties

(a) False for ordinary homology: a nonempty contractible space has $H_0 \cong \mathbb{Z}$. Its positive-dimensional groups vanish.

(b) True. A continuous map induces homomorphisms in each dimension; on simplicial chains the construction sends cycles to cycles and boundaries to boundaries.

(c) True. Each is a quotient of abelian groups.

(d) False. A Klein bottle has a $\mathbb{Z}_2$ torsion summand in H_1, which cannot occur in a free abelian group.

(e) True. H_0 is free abelian on the connected components (for a finite triangulation, one copy of $\mathbb{Z}$ per component).

(f) True. If there are no $(n+1)$-simplexes, $B_n = 0$ and $H_n = Z_n \leq C_n$. A subgroup of a free abelian group is free abelian.

(g) True by the definition of the boundary map.

(h) True because applying one more boundary gives zero.

(i) True because $\partial^2 = 0$.

(j) False. A group with torsion, such as H_1 of the Klein bottle, cannot be a subgroup of the torsion-free group C_1. In general homology classes are cosets, not distinguished chains.

Exercise 43.7: Euler characteristic of a surface with handles

For the sphere with n handles, Exercise 42.12 gives ranks $b_0 = 1$, $b_1 = 2n$, $b_2 = 1$, and all others zero. The Euler–Poincaré formula therefore gives

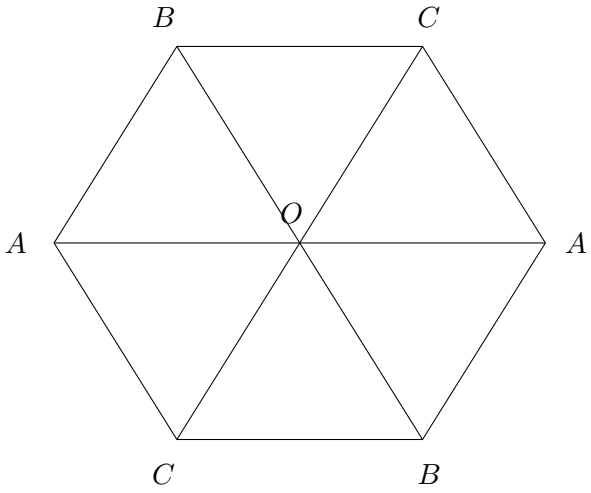
$$\chi = 1 - 2n + 1 = 2 - 2n.$$

This equals 2 for the sphere, 0 for the torus, and decreases by two for every added handle. A polygon cell model also gives the same count: one vertex, $2n$ edges, and one face.

Exercise 43.8: Triangulating and computing the projective plane

Build a genuine triangulation from the disk model.

Replace the circular boundary by a hexagon, keeping the antipodal identification. The paired boundary vertices have labels A, B, C, A, B, C in order; join all six to an interior vertex O.



Before identifying the boundary, barycentrically subdivide each of the six triangles: place a midpoint on each edge, a center in each triangle, and draw the six small triangles corresponding to vertex–edge–triangle flags. Now identify opposite boundary vertices and edge pieces. The subdivision prevents different triangles from having the same vertex set after gluing, so this gives an actual simplicial complex rather than a polygon with folded edges. It has 19 vertices, 54 edges, and 36 triangular faces, with Euler characteristic 1.

Compute the cycles and relations.

Eliminate interior edges using boundaries of the small triangles, just as in the torus calculation. All 1-cycles reduce to multiples of one loop a along the identified semicircles. Traversing the disk boundary goes around a twice in the same direction, so the remaining face boundary is $2a$. Equivalently, the reduced chain calculation has one vertex, one edge a, and one face F, with

$$\partial_1 a = 0, \qquad \partial_2 F = 2a.$$

The elimination consists of choosing and removing interior tree edges and cancelling adjacent face–edge pairs, so it leaves homology unchanged. Thus no nonzero multiple of F is a 2-cycle, while the dimension-one relation is exactly $2a = 0$. The space is connected. Therefore

$$H_0(\mathbb{RP}^2) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^2) \cong \mathbb{Z}_2, \quad H_n(\mathbb{RP}^2) = 0 \ (n \geq 2).$$

The factor 2 comes from the reversed edge gluing, not from a count of two independent loops.

Exercise 43.9: A sphere with q cross caps

For $q \geq 1$ the polygon model has one vertex, loops $a_1, \ldots, a_q$, and one face whose boundary word is $a_1^2 \cdots a_q^2$. Triangulate the polygon and eliminate interior edges as in Exercise 43.8. The resulting

chain maps are

$$\partial_1 = 0, \qquad \partial_2(1) = 2(a_1 + \cdots + a_q).$$

This map $\mathbb{Z} \to \mathbb{Z}^q$ is injective, so $H_2 = 0$. The vector $u = a_1 + \cdots + a_q$, together with $a_2, \ldots, a_q$, is an integer basis: solve $a_1 = u - a_2 - \cdots - a_q$. In this basis the single relation is $2u = 0$. Hence

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}^{q-1} \oplus \mathbb{Z}_2, \qquad H_n = 0 \ (n \geq 2).$$

For $q = 1$ this recovers the projective plane; for $q = 2$ it recovers the Klein bottle. With no cross caps, $q = 0$, the space is the sphere instead, with $H_2 = \mathbb{Z}$ and $H_1 = 0$.

Exercise 43.10: Induced maps of torus coverings

In the figure, varying ϕ gives the meridian a, while varying θ gives the longitude b. Use $([a], [b])$ as the ordered basis of $H_1 \cong \mathbb{Z}^2$ and the oriented whole torus $[T]$ as the generator of $H_2 \cong \mathbb{Z}$. In every case $H_0 \cong \mathbb{Z}$ is sent by the identity, since a point maps to a point in a connected space.

(a) For $(\theta, \phi) \mapsto (2\theta, \phi)$, the meridian wraps once and the longitude twice: $[a] \mapsto [a]$, $[b] \mapsto 2[b]$. Thus the H_1 matrix is $\mathrm{diag}(1, 2)$. The whole oriented surface wraps twice, so $[T] \mapsto 2[T]$.

(b) For $(\theta, \phi) \mapsto (\theta, 2\phi)$, $[a] \mapsto 2[a]$, $[b] \mapsto [b]$. The H_1 matrix is $\mathrm{diag}(2, 1)$ and $[T] \mapsto 2[T]$.

(c) For doubling both angles, $[a] \mapsto 2[a]$, $[b] \mapsto 2[b]$. The H_1 matrix is $2I$, and the surface wraps twice in each direction, giving $[T] \mapsto 4[T]$.

The dimension-two multipliers are the determinants of the dimension-one matrices; the positive signs agree with preservation of orientation in each angular direction.

Exercise 43.11: Maps from the torus onto its longitude

The target circle b has $H_0(b) = \mathbb{Z}$, $H_1(b) = \mathbb{Z}[b]$, and $H_2(b) = 0$. Each map sends the H_0 point generator to the point generator, so its H_0 map is the identity. **(a)** For $f(\theta, \phi) = (\theta, 0)$, a meridian becomes a point and the longitude goes around once. Hence $f_*[a] = 0$, $f_*[b] = [b]$, or $(r, s) \mapsto s$ on H_1. **(b)** For $f(\theta, \phi) = (2\theta, 0)$, the same meridian collapses and the longitude goes around twice. Thus $(r, s) \mapsto 2s$ on H_1. In both cases the H_2 map is zero because its target group is zero. Geometrically no two-dimensional surface class can survive a map whose image is a circle.

Exercise 43.12: The same maps viewed as torus self-maps

Now the target has $H_1 = \mathbb{Z}[a] \oplus \mathbb{Z}[b]$ and $H_2 = \mathbb{Z}[T]$, but the maps still factor through the circle b. For part **(a)**, $[a] \mapsto 0$ and $[b] \mapsto [b]$, so the H_1 matrix is $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. For part **(b)**, $[a] \mapsto 0$ and $[b] \mapsto 2[b]$, so it is $\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$. Both maps induce the identity on H_0. Both induce zero on H_2, even though the target H_2 is nonzero: factorization through b gives a factor through $H_2(b) = 0$. The determinant-zero H_1 matrices provide a consistency check.

Exercise 43.13: Projecting the Klein bottle onto b

The Klein bottle has $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}_2[a] \oplus \mathbb{Z}[b]$, and $H_2 = 0$ in the conventions of Fig. 43.2. The projection to the vertical circle b collapses each horizontal a loop to a point and carries b onto itself once. Therefore the induced maps are

$$H_0 :\ m \mapsto m, \qquad H_1 :\ (\overline{r}, s) \mapsto s, \qquad H_2 :\ 0 \longrightarrow 0.$$

The dimension-one formula is well defined on the order-2 class of a, since that class maps to zero. It also follows algebraically that any homomorphism from its $\mathbb{Z}_2$ summand to the torsion-free group $\mathbb{Z}$ must vanish.